

Gradient-Based Control of a Chaotic System: The Lyapunov-Time Barrier in Differentiable Simulation

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Abstract

We study the problem of learning a state-feedback controller for the Lorenz system by backpropagating through a differentiable simulator. The control objective is to confine the trajectory to a single lobe of the attractor. We develop the theory of the sensitivity $\partial \mathbf{s}(T)/\partial u$ for a chaotic plant and show that it grows like $e^{\lambda_1 T}$, so that the gradient of any trajectory-averaged loss inherits the same exponential growth in the length of the training window. This gives a hard horizon — measured in Lyapunov times — beyond which gradient descent through the simulator ceases to be a usable optimization method, not because the gradient is wrong but because it is enormous, stiff, and supported on an increasingly fractal loss landscape. We confirm the prediction numerically: the loss gradient grows at 0.926 nats per Lyapunov time against a theoretical rate of 1, spanning 11.4 decades over a 25τ window. We then analyse an L^2 control-effort regularizer, and show by closed-loop equilibrium analysis that the learned linear feedback law solves the task by manufacturing a *stable* fixed point out of what is an unstable spiral saddle in the free system. The regularizer does not merely reduce actuation cost: it changes which invariant set the closed loop settles onto, replacing a violent limit cycle with a quiet equilibrium at less than half the RMS control effort.

Contents

1	Introduction	2
2	The plant	3
2.1	Equations and parameters	3
2.2	Equilibria of the free system	3
2.3	The Lyapunov time as the natural clock	3
2.4	Discretization	4
3	Actuation and the control law	4
4	The objective	4
4.1	A smooth lobe indicator	4
4.2	Task loss	5
4.3	Control effort	5
4.4	Composite objective	5
5	Differentiable simulation and the discrete adjoint	5
5.1	The computational graph	5
5.2	The adjoint recursion	6
5.3	The identity that causes the problem	6

6	The Lyapunov-time barrier	6
6.1	Growth law	6
6.2	Two distinct horizons	7
6.3	Numerical confirmation	8
7	Control-effort regularization	9
7.1	Why the unregularized problem is ill-posed	9
7.2	Effect on conditioning	9
7.3	Choosing λ	9
8	What the learned controller does	9
8.1	Closed-loop equilibria	10
8.2	The learned controllers	10
8.3	Equilibrium versus limit cycle	10
9	Numerical results	11
9.1	Protocol	11
9.2	The barrier in the optimizer	11
9.3	Generalization beyond the training window	12
10	Discussion	12
10.1	Summary	12
10.2	Remedies	13
10.3	Limitations	13
A	Notation	14
B	Correspondence with the implementation	14

1 Introduction

Differentiable simulation — writing a physical model in an automatic differentiation framework and optimizing its parameters or its controller by gradient descent — has become a default tool. It is spectacularly effective for dissipative, contracting, or short-horizon problems. This document is concerned with the regime where it fails, and with characterising the failure precisely enough that the boundary can be predicted in advance rather than discovered by experiment.

The failure mode is specific to chaotic plants. In a chaotic system nearby trajectories separate exponentially, and the same linear machinery that produces that separation — the tangent propagator — is exactly the object that reverse-mode automatic differentiation accumulates when it backpropagates through the time loop. Sensitivity and instability are two readings of the same matrix product. One cannot have a system that is interesting to control and a gradient that is well behaved over long horizons; they are the same phenomenon.

We make this concrete on the Lorenz system with a single scalar actuator, a linear feedback law, and a smooth surrogate for the discrete objective “stay in the right-hand lobe.” Sections 2–4 set up the plant, the actuation model and the loss. Section 5 derives the discrete adjoint recursion that backpropagation implements. Section 6 contains the main theoretical content: the exponential growth law for the loss gradient, the resulting horizon, and its numerical confirmation. Section 7 treats the control-effort regularizer. Section 8 analyses what the learned controller actually does, via the equilibria and Jacobian spectrum of the closed loop. Section 9 collects the numerical results, and Section 10 discusses remedies.

2 The plant

2.1 Equations and parameters

The Lorenz system is the three-dimensional autonomous flow

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(\rho - z) - y, \\ \dot{z} &= xy - \beta z,\end{aligned}\tag{1}$$

with the classical parameter values used throughout this work,

$$\sigma = 10, \quad \rho = 28, \quad \beta = \frac{8}{3}.\tag{2}$$

Write $\mathbf{s} = (x, y, z)^\top \in \mathbb{R}^3$ and $f(\mathbf{s})$ for the right-hand side of (1), so that $\dot{\mathbf{s}} = f(\mathbf{s})$.

The flow is dissipative: $\nabla \cdot f = -(\sigma + 1 + \beta) \approx -13.67 < 0$, so phase-space volume contracts uniformly at rate $e^{-13.67t}$. All trajectories are therefore eventually confined to a set of zero volume — the attractor — on which the dynamics is nonetheless chaotic. Contraction of volume and separation of nearby points coexist because the contraction is anisotropic.

2.2 Equilibria of the free system

Setting $f(\mathbf{s}) = 0$ gives three equilibria. The origin $\mathbf{s}_0 = (0, 0, 0)$ and the symmetric pair

$$\mathbf{s}_\pm = \left(\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1 \right) = (\pm 8.485, \pm 8.485, 27).\tag{3}$$

The Jacobian of (1) is

$$\mathbf{J}(\mathbf{s}) = \frac{\partial f}{\partial \mathbf{s}} = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix}.\tag{4}$$

At the origin the spectrum is $\{-22.83, 11.83, -2.67\}$: a saddle with a one-dimensional unstable manifold. At $\mathbf{s}_\pm$ the spectrum is $\{-13.855, 0.094 \pm 10.195i\}$. The complex pair has *positive* real part, so $\mathbf{s}_\pm$ are unstable spiral saddles: trajectories spiral outward from each lobe centre until they are ejected and captured by the other lobe. This ejection-and-capture cycle is precisely the lobe switching that the controller of this work is asked to suppress. The margin of instability is small — $\text{Re } \mu = 0.094$ — which is why a modest control authority suffices to reverse it (Section 8).

2.3 The Lyapunov time as the natural clock

Let $\delta\mathbf{s}(t)$ be an infinitesimal perturbation to a trajectory. It obeys the variational equation

$$\dot{\delta\mathbf{s}} = \mathbf{J}(\mathbf{s}(t)) \delta\mathbf{s}, \quad \delta\mathbf{s}(t) = \Phi(t, 0) \delta\mathbf{s}(0),\tag{5}$$

where $\Phi(t, s)$ is the state-transition (tangent propagator) matrix. The maximal Lyapunov exponent is

$$\lambda_1 = \lim_{t \rightarrow \infty} \frac{1}{t} \log \|\Phi(t, 0)\|,\tag{6}$$

and for (1) at the classical parameters

$$\lambda_1 \approx 0.9056, \quad \tau \equiv \lambda_1^{-1} \approx 1.104,\tag{7}$$

τ being the *Lyapunov time*. Throughout this document, horizons are reported in Lyapunov times $t/\tau = \lambda_1 t$, which is the only dimensionless clock that makes statements about predictability parameter-independent. One Lyapunov time is the interval over which an infinitesimal uncertainty is amplified by a factor of e .

2.4 Discretization

The simulator integrates (1) with explicit Euler at fixed step $\Delta t = 0.01$:

$$\mathbf{s}_{k+1} = F(\mathbf{s}_k, u_k) \equiv \mathbf{s}_k + \Delta t [f(\mathbf{s}_k) + u_k \mathbf{e}_1], \quad \mathbf{e}_1 = (1, 0, 0)^\top. \quad (8)$$

A training window of L Lyapunov times corresponds to

$$N = \text{round}\left(\frac{L}{\lambda_1 \Delta t}\right) \approx 110 L \text{ steps}. \quad (9)$$

Euler at $\Delta t = 0.01$ is well inside the stability region for the fastest eigenvalue ($|\lambda| \Delta t \approx 0.23$ at the origin) and is used here for the transparency of its adjoint rather than for accuracy; the $O(\Delta t)$ local truncation error is irrelevant to the argument of Section 6, which concerns the amplification of *any* perturbation, including discretization error.

3 Actuation and the control law

We give the system a single scalar actuator entering the x -channel:

$$\begin{aligned} \dot{x} &= \sigma(y - x) + u, \\ \dot{y} &= x(\rho - z) - y, \\ \dot{z} &= xy - \beta z. \end{aligned} \quad (10)$$

Physically, in the Rayleigh–Bénard reading of the Lorenz model, x is the amplitude of the convective roll and u is a forcing applied directly to that mode.

The controller is the simplest object that can close the loop: an affine state-feedback law with three weights and a bias,

$$u_k = \pi_\theta(\mathbf{s}_k) = \mathbf{w}^\top \mathbf{s}_k + b, \quad \theta = (\mathbf{w}, b) \in \mathbb{R}^4. \quad (11)$$

Two properties make this choice deliberate rather than merely convenient. First, the parameter count is small enough that the entire loss landscape is attributable to the dynamics rather than to network overparameterization — any pathology observed is a property of the plant, not of the function approximator. Second, the closed loop $\dot{\mathbf{s}} = f(\mathbf{s}) + (\mathbf{w}^\top \mathbf{s} + b) \mathbf{e}_1$ remains a polynomial vector field of the same degree as (1), so its equilibria and their stability can be computed in closed form. This is what makes Section 8 possible: we can say exactly what the learned controller did, not merely that it worked.

Note that u is unbounded — there is no saturation. The consequences are taken up in Section 7.

4 The objective

4.1 A smooth lobe indicator

The task, “keep the trajectory in the right-hand lobe,” is the statement $x(t) > 0$ for all t . The natural loss, the fraction of time spent with $x \leq 0$, is piecewise constant in θ and has zero gradient almost everywhere. We therefore replace the indicator $\mathbf{1}[x > 0]$ by a smooth surrogate,

$$\varphi_\varepsilon(x) = \frac{1}{2} \left(1 + \tanh \frac{x}{\varepsilon} \right) \in (0, 1), \quad \varepsilon = 2. \quad (12)$$

This is the logistic sigmoid $\sigma_L(2x/\varepsilon)$ in disguise. It satisfies $\varphi_\varepsilon \rightarrow \mathbf{1}[x > 0]$ pointwise as $\varepsilon \downarrow 0$, and its derivative

$$\varphi'_\varepsilon(x) = \frac{1}{2\varepsilon} \text{sech}^2 \frac{x}{\varepsilon} \quad (13)$$

is concentrated in a band of width $O(\varepsilon)$ about the separating plane $x = 0$. The parameter ε therefore sets how far from the lobe boundary the loss can still “see” the trajectory. Taking ε too small recovers the vanishing-gradient problem of the hard indicator; taking it too large makes the loss insensitive to the actual lobe occupancy. With $\varepsilon = 2$ against a lobe half-width of ≈ 8.5 , the transition band occupies roughly a quarter of the lobe — a compromise that keeps gradient signal available while the trajectory is still well inside the wrong lobe.

4.2 Task loss

For a discrete trajectory $P = (\mathbf{s}_0, \dots, \mathbf{s}_N)$,

$$\mathcal{L}_{\text{task}}(P) = \frac{1}{N+1} \sum_{k=0}^N [1 - \varphi_\varepsilon(x_k)] \approx \frac{1}{T} \int_0^T [1 - \varphi_\varepsilon(x(t))] dt, \quad (14)$$

a soft measure of the fraction of the window spent in the left lobe. It is bounded in $(0, 1)$, equals $\frac{1}{2}$ for a trajectory pinned at $x = 0$, and is minimized by driving x positive and large.

4.3 Control effort

Left alone, (14) is indifferent to how much actuation is spent. With an unbounded actuator this is an ill-posed request: the infimum is approached by ever more aggressive laws. We add a quadratic penalty on the applied control,

$$\mathcal{L}_{\text{eff}}(P, \theta) = \frac{1}{N} \sum_{k=0}^{N-1} \left(\frac{u_k}{u_{\text{ref}}} \right)^2, \quad u_{\text{ref}} = 60. \quad (15)$$

Two details matter. The sum runs to $N - 1$: the control evaluated at the terminal state $\mathbf{s}_N$ is never applied to anything and must not be charged for, or the penalty acquires a spurious terminal term whose gradient is the most ill-conditioned in the whole window. And the normalization by u_{ref} makes $\mathcal{L}_{\text{eff}}$ dimensionless and $O(1)$ at the actuation scale of interest, so that the trade-off weight λ is comparable across problems: $\mathcal{L}_{\text{eff}} = 1$ means an RMS control of u_{ref} . The quantity actually reported during training is the physical RMS, $\|u\|_{\text{rms}} = u_{\text{ref}} \sqrt{\mathcal{L}_{\text{eff}}}$.

4.4 Composite objective

$$\boxed{\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(P_\theta) + \lambda \mathcal{L}_{\text{eff}}(P_\theta, \theta), \quad \lambda = 0.007} \quad (16)$$

where P_θ is the closed-loop trajectory generated by (8) and (11) from a fixed initial condition. Setting $\lambda = 0$ recovers the unregularized problem. The value $\lambda = 0.007$ is chosen so that an RMS control of $u_{\text{ref}} = 60$ incurs a penalty of 0.007, about 1.5% of the task loss of a trajectory that ignores the objective entirely ($\mathcal{L}_{\text{task}} \approx 0.5$). The regularizer is thus deliberately weak: it is a tie-breaker among controllers that solve the task, not a competing objective.

5 Differentiable simulation and the discrete adjoint

5.1 The computational graph

The simulator is written so that the entire rollout is a differentiable composition

$$\theta \mapsto \mathbf{s}_1 \mapsto \mathbf{s}_2 \mapsto \dots \mapsto \mathbf{s}_N \mapsto \mathcal{L}, \quad (17)$$

with $\mathbf{s}_{k+1} = F(\mathbf{s}_k, \pi_\theta(\mathbf{s}_k))$. Reverse-mode automatic differentiation over this graph is backpropagation through time (BPTT), and it is numerically identical to the discrete adjoint method. Making that identity explicit is what exposes the pathology.

5.2 The adjoint recursion

Define the closed-loop one-step Jacobian

$$A_k \equiv \frac{\partial \mathbf{s}_{k+1}}{\partial \mathbf{s}_k} = I + \Delta t \left[\mathbf{J}(\mathbf{s}_k) + e_1 \mathbf{w}^\top \right], \quad B_k \equiv \frac{\partial \mathbf{s}_{k+1}}{\partial u_k} = \Delta t e_1, \quad (18)$$

where $\mathbf{J}$ is (4) and the rank-one term $e_1 \mathbf{w}^\top$ is the contribution of the feedback law. Let ℓ_k denote the per-step contribution to $\mathcal{L}$, and define the adjoint (costate) $a_k = \partial \mathcal{L} / \partial \mathbf{s}_k \in \mathbb{R}^{1 \times 3}$. Then

$$a_N = \frac{\partial \ell_N}{\partial \mathbf{s}_N}, \quad a_k = \frac{\partial \ell_k}{\partial \mathbf{s}_k} + a_{k+1} \left(A_k + B_k \frac{\partial \pi_\theta}{\partial \mathbf{s}_k} \right), \quad (19)$$

integrated backwards from $k = N - 1$ to 0, and the parameter gradient accumulates as

$$\nabla_\theta \mathcal{L} = \sum_{k=0}^{N-1} \left(a_{k+1} B_k + \lambda \frac{\partial \mathcal{L}_{\text{eff}}}{\partial u_k} \right) \frac{\partial \pi_\theta(\mathbf{s}_k)}{\partial \theta}, \quad \frac{\partial \pi_\theta}{\partial \theta} = (\mathbf{s}_k^\top, 1). \quad (20)$$

Equation (19) is a linear recursion driven backwards by products of the matrices A_k . Its solution is

$$a_k = \sum_{j \geq k} \frac{\partial \ell_j}{\partial \mathbf{s}_j} \Phi_{k \rightarrow j}, \quad \Phi_{k \rightarrow j} = \tilde{A}_{j-1} \tilde{A}_{j-2} \cdots \tilde{A}_k, \quad (21)$$

with $\tilde{A}_k = A_k + B_k \partial_{\mathbf{s}} \pi_\theta$ the closed-loop transition matrix. The discrete product $\Phi_{k \rightarrow j}$ is exactly the discretized tangent propagator $\Phi(t_j, t_k)$ of (5), for the closed-loop vector field.

5.3 The identity that causes the problem

Proposition 1 (Sensitivity is the tangent propagator). *The sensitivity of the loss to the controller parameters is a Φ -weighted sum of per-step loss sensitivities, where Φ is the same linear operator whose growth rate defines the Lyapunov exponent of the closed loop.*

This is immediate from (21), and it is the entire difficulty. There is no way to compute a long-horizon gradient of a chaotic simulation that does not, somewhere, form the product $\tilde{A}_{N-1} \cdots \tilde{A}_0$. If the closed loop is chaotic — which it is, until the controller has already succeeded — that product has norm $\sim e^{\lambda_1 T}$. Sensitivity analysis and instability analysis are the same calculation read in two directions.

Remark 1. *The same statement holds for the open-loop sensitivity to a constant control. The routine `position_gradient` in the accompanying code measures $\partial x(T) / \partial u$ for a constant u held over $[0, T]$, which by (21) equals $\Delta t \sum_k e_1^\top \Phi_{k \rightarrow N} e_1$ — a sum of N terms whose last is $O(1)$ and whose first is $O(e^{\lambda_1 T})$. The sum is dominated by its earliest term.*

6 The Lyapunov-time barrier

6.1 Growth law

Proposition 2 (Exponential gradient growth). *Let $\mathcal{L}_T$ be a trajectory-averaged loss over a window of length T , and suppose the closed loop has maximal Lyapunov exponent $\lambda_1 > 0$ with a generic (non-degenerate) projection of the loss gradient onto the unstable subspace. Then*

$$\|\nabla_\theta \mathcal{L}_T\| = \Theta\left(\frac{e^{\lambda_1 T}}{T}\right) \sim C e^{T/\tau} \quad \text{up to the algebraic prefactor.} \quad (22)$$

Sketch. By (20) and (21), the gradient is a sum over k of terms $a_{k+1}B_k\partial_\theta\pi$ where $a_k = \sum_{j \geq k}(\partial\ell_j/\partial\mathbf{s}_j)\Phi_{k \rightarrow j}$. By Oseledets' theorem $\|\Phi_{k \rightarrow j}\| \asymp e^{\lambda_1(t_j - t_k)}$ along almost every orbit. The dominant contribution is $k = 0$, $j = N$, giving $e^{\lambda_1 T}$; the $1/N$ from the trajectory average supplies the $1/T$; and the sum of the remaining terms is a geometric series dominated by its largest element, contributing only a constant factor. Non-degeneracy is needed to exclude the measure-zero case in which $\partial\ell/\partial\mathbf{s}$ is orthogonal to the leading Oseledets direction for every j . $\square$

Rewriting (22) in Lyapunov times $L = T/\tau$,

$$\log_{10}\|\nabla_\theta\mathcal{L}\| \approx \frac{L}{\ln 10} + \text{const} \approx 0.434 L + \text{const}, \quad (23)$$

so the gradient gains roughly one order of magnitude every 2.3 Lyapunov times. This is the quantitative statement that the experiment of Section 6.3 tests.

6.2 Two distinct horizons

It is worth separating two limits that are often conflated.

The arithmetic horizon. Relative rounding error $\epsilon_m = 2.2 \times 10^{-16}$ (IEEE double) is injected at every step and amplified by $e^{\lambda_1 t}$. The computed trajectory ceases to shadow the true one — and the computed gradient ceases to have any correct digits — at

$$L_{\text{arith}} \approx \ln\left(\frac{1}{\epsilon_m}\right) \approx 36 \text{ Lyapunov times}, \quad (24)$$

using $L = \lambda_1 T$ so that the exponent is measured in nats per Lyapunov time. In single precision this drops to $L_{\text{arith}} \approx 16$. This is the horizon usually quoted, and it is generous.

The optimization horizon. Long before arithmetic fails, the gradient becomes useless as a *search direction*. Three effects compound:

1. *Scale.* With $\|\nabla\mathcal{L}\| \sim 10^9$ and a bounded loss $\mathcal{L} \in (0, 1)$, the linearization $\mathcal{L}(\theta + \delta) \approx \mathcal{L}(\theta) + \nabla\mathcal{L} \cdot \delta$ is valid only for $\|\delta\| \lesssim 10^{-9}$. The trust region collapses at the same exponential rate as the gradient grows.
2. *Landscape roughness.* The map $\theta \mapsto \mathcal{L}_T(\theta)$ inherits the fractal structure of the attractor's basin boundaries. As T grows, the set of θ for which the trajectory switches lobes at least once develops structure on scales $\sim e^{-\lambda_1 T}$. The loss is still differentiable at each point, but its gradient becomes uninformative about behaviour at any finite step size — the descent direction at θ says nothing about $\mathcal{L}$ at $\theta + \delta$ for any δ a practical optimizer would take.
3. *Adaptive-optimizer masking.* Adam normalizes by a running second-moment estimate and so is invariant to the *overall* scale of the gradient. This is why training does not visibly diverge — and it is precisely what makes the failure insidious. Adam removes the symptom (huge steps) while leaving the disease (a direction that is dominated by the projection onto the leading unstable Oseledets direction of a single early step, and that carries no information about the average behaviour the loss is supposed to measure).

Empirically (Section 9) the optimization horizon for this problem lies between 1 and 2 Lyapunov times — more than an order of magnitude below (24). *The binding constraint on differentiable simulation of chaos is not floating point. It is that the gradient stops being a useful descent direction while every digit of it is still correct.*

6.3 Numerical confirmation

To measure (22) free of any optimizer, we sample training windows $L \in [0.1, 25]$, and for each one instantiate a *fresh* zero-initialized controller $\theta = 0$ (so the closed loop is the free Lorenz system and λ_1 is the plant’s own exponent), roll out $N = 110L$ steps, and record $\|\nabla_{\theta} \mathcal{L}_{\text{task}}\|_2$.

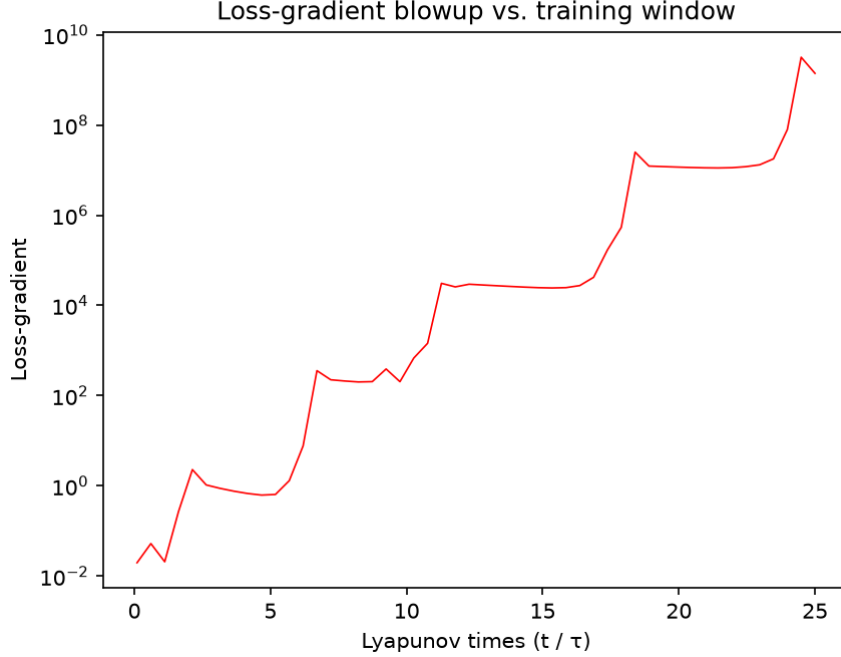


Figure 1: Norm of the loss gradient with respect to the controller parameters as a function of the training window, on a logarithmic ordinate. Each point is an independent rollout from the same initial condition with $\theta = 0$. The gradient spans 11.4 decades over 25 Lyapunov times. The staircase profile — plateaus punctuated by near-vertical jumps — is the signature of a trajectory that accumulates most of its sensitivity during discrete lobe transitions, when the orbit passes near the origin’s stable manifold and the local expansion rate far exceeds its long-time average λ_1 .

Figure 1 shows the result. A least-squares fit of $\log \|\nabla \mathcal{L}\|$ against L over 60 windows gives

$$\frac{d}{dL} \log \|\nabla_{\theta} \mathcal{L}\| = 0.926 \text{ nats per Lyapunov time}, \quad (25)$$

against the theoretical prediction of exactly 1 from (22). Equivalently the measured growth exponent is $\lambda_{\text{eff}} = 0.926 \lambda_1 = 0.839$, within 7% of the independently known $\lambda_1 = 0.9056$. The residual shortfall is expected: the fit is over a single finite orbit, so it estimates a finite-time Lyapunov exponent, and the $1/T$ prefactor in (22) biases the slope slightly downward.

The staircase structure deserves comment because it is not noise. Between lobe transitions the orbit spirals within a lobe, where the local expansion is close to neutral and the gradient plateaus; each near-passage of the origin’s stable manifold contributes a burst of expansion far above λ_1 . The long-time exponent λ_1 is an average over these bursts, so a controller trained on a window that happens to end on a plateau will see a very different gradient magnitude from one whose window ends just past a burst — an additional source of variance in the optimization that the smooth law (22) hides.

7 Control-effort regularization

7.1 Why the unregularized problem is ill-posed

With $\lambda = 0$ and an unbounded actuator, consider the scaling $\theta \mapsto c\theta$ for large c . The control $u = c(\mathbf{w}^\top \mathbf{s} + b)$ dominates the $O(\rho x)$ terms of the plant, and in the limit the x -equation becomes a fast linear system that can hold x at any desired value. The task loss $\mathcal{L}_{\text{task}}$ can thus be driven arbitrarily close to its infimum along a direction in which $\|\theta\| \rightarrow \infty$. The minimizer does not exist; only a minimizing sequence does. Gradient descent from a finite initialization does not run off to infinity in practice, but it does drift along this flat, unbounded valley, and where it stops is determined by the iteration budget rather than by the objective.

Adding $\lambda \mathcal{L}_{\text{eff}}$ with $\lambda > 0$ makes $\mathcal{L}$ coercive in θ — $\mathcal{L}(\theta) \rightarrow \infty$ as $\|\theta\| \rightarrow \infty$ whenever the trajectory does not collapse to a point — so sublevel sets are bounded and minimizers exist. This is the well-posedness argument, and it is independent of any claim about energy cost.

7.2 Effect on conditioning

The penalty term contributes a directly computable positive-semidefinite piece to the Hessian. Writing $\mathcal{L}_{\text{eff}} = \frac{1}{Nu_{\text{ref}}^2} \sum_k (\mathbf{w}^\top \mathbf{s}_k + b)^2$ and holding the trajectory fixed,

$$\nabla_{\theta}^2 \mathcal{L}_{\text{eff}} \Big|_{P \text{ fixed}} = \frac{2}{Nu_{\text{ref}}^2} \sum_{k=0}^{N-1} \begin{pmatrix} \mathbf{s}_k \\ 1 \end{pmatrix} \begin{pmatrix} \mathbf{s}_k \\ 1 \end{pmatrix}^\top = \frac{2}{u_{\text{ref}}^2} \widehat{M}, \quad (26)$$

with $\widehat{M}$ the second-moment matrix of the augmented state along the trajectory. Since the Lorenz attractor spans all three coordinates, $\widehat{M} \succ 0$, and the penalty adds a strictly positive floor of order $2\lambda\sigma_{\min}(\widehat{M})/u_{\text{ref}}^2$ to every eigenvalue of the Hessian.

This is a genuine but bounded improvement, and it should not be oversold. The task term’s Hessian carries entries of order $e^{2\lambda_1 T}$ from the squared propagator; the additive floor (26) is $O(\lambda)$ and independent of T . The condition number therefore still degrades like $e^{2\lambda_1 T}$. *Regularization does not defeat the Lyapunov barrier; it removes a degeneracy that is orthogonal to it.* The barrier is a property of the plant and is not removable by any choice of penalty on θ .

7.3 Choosing λ

The trade-off is between the task loss and the actuation cost. Because $\mathcal{L}_{\text{eff}}$ is normalized by u_{ref} , the two terms are commensurate when $\lambda \approx \mathcal{L}_{\text{task}}$ at the target RMS, which for $\mathcal{L}_{\text{task}} \sim 10^{-2}$ and $\|u\|_{\text{rms}} \sim u_{\text{ref}}$ gives $\lambda \sim 10^{-2}$. The value used, $\lambda = 0.007$, sits just below this: strong enough to select among task-solving controllers, weak enough not to trade task performance for quiet. Section 8 shows that at this setting the regularizer improves *both* objectives simultaneously — which is possible only because the unregularized problem was not converging to a genuine optimum of the task in the first place.

8 What the learned controller does

The virtue of the affine law (11) is that the closed loop can be analysed exactly. This section computes the equilibria of the controlled system and their stability, and shows that the learned controllers solve the task by a mechanism that the loss never mentions.

8.1 Closed-loop equilibria

Substituting $u = \mathbf{w}^\top \mathbf{s} + b$ into (10) and setting $\dot{\mathbf{s}} = 0$: from $\dot{z} = 0$, $z = xy/\beta$; substituting into $\dot{y} = 0$ gives

$$y(x) = \frac{\rho\beta x}{\beta + x^2}, \quad z(x) = \frac{\rho x^2}{\beta + x^2}, \quad (27)$$

which are exactly the free-system relations, since the actuator touches only the x -equation. Substituting (27) into $\dot{x} = 0$ and clearing the denominator yields a scalar cubic in x :

Proposition 3 (Equilibrium cubic). *The equilibria of the closed loop are $(x_*, y(x_*), z(x_*))$ where x_* solves*

$$(w_1 - \sigma)x^3 + (b + \rho w_3)x^2 + \beta[(w_1 - \sigma) + \rho(\sigma + w_2)]x + \beta b = 0. \quad (28)$$

Setting $\mathbf{w} = 0, b = 0$ reduces (28) to $-\sigma x[x^2 - \beta(\rho - 1)] = 0$, recovering the origin and $\mathbf{s}_\pm$ of (3), as it must. The controller has four parameters and the cubic has three coefficients that they can move independently (the constant and cubic coefficients are coupled through b and w_1), so the learned law has enough freedom to place an equilibrium essentially anywhere on the curve (27).

Stability is read off the closed-loop Jacobian, which is (4) plus the rank-one feedback term:

$$\tilde{\mathbf{J}}(\mathbf{s}) = \mathbf{J}(\mathbf{s}) + \mathbf{e}_1 \mathbf{w}^\top = \begin{pmatrix} -\sigma + w_1 & \sigma + w_2 & w_3 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix}. \quad (29)$$

8.2 The learned controllers

Table 1 applies (28) and (29) to the controllers learned at a training window of one Lyapunov time. Both create a stable equilibrium in the right-hand lobe. In the free system, the corresponding point $\mathbf{s}_+$ has eigenvalues $-13.855, 0.094 \pm 10.195i$ — an unstable spiral. Both controllers push that unstable complex pair across the imaginary axis: the regularized law achieves $\text{Re } \mu = -0.891$, a swing of 0.985 in the real part, at the cost of raising the oscillation frequency from 10.2 to 12.8.

Table 1: Equilibria and spectra in the right lobe. The controllers are those learned with a 1τ training window (Table 2). Both convert the unstable spiral $\mathbf{s}_+$ into an attracting focus.

	Free system	Unregularized	Regularized
Equilibrium (x, y, z)	(8.485, 8.485, 27.000)	(18.972, 3.907, 27.794)	(13.083, 5.619, 27.570)
Real eigenvalue	-13.855	-13.135	-15.054
Complex pair	$+0.094 \pm 10.195i$	$-1.213 \pm 18.551i$	$-0.891 \pm 12.830i$
max $\text{Re } \mu$	+0.094 (unstable)	-1.213 (stable)	-0.891 (stable)

Both closed loops also retain a saddle near the origin ($\max \text{Re } \mu = +3.58$ regularized, $+4.07$ unregularized) and acquire a third stable equilibrium deep in the left half-space. The controller does not — and is not asked to — render the right-lobe equilibrium globally attracting. It only has to make its basin contain the initial condition.

8.3 Equilibrium versus limit cycle

Local stability is necessary but not sufficient, and here the two controllers part company. Integrating both closed loops for 50 Lyapunov times from the training initial condition $(0, 1, 1.05)$:

- The **regularized** controller converges to its equilibrium. Over the last quarter of the run, $x \in [13.083, 13.084]$ and the control settles to a constant $u = 74.6$. The decay rate $e^{-0.891t}$ predicted by Table 1 gives a settling time of ≈ 5 time units per e -folding, consistent with the envelope visible in Figure 2b.

- The **unregularized** controller does *not* converge to its equilibrium, despite that equilibrium being more strongly stable ($\text{Re } \mu = -1.213$). It settles instead onto a large-amplitude attracting limit cycle with $x \in [1.41, 29.78]$ and $|u|$ reaching 435.8, at an RMS effort of 164.9 — more than twice the regularized law’s 74.6. The initial condition lies outside the equilibrium’s basin, which the aggressive feedback has made small.

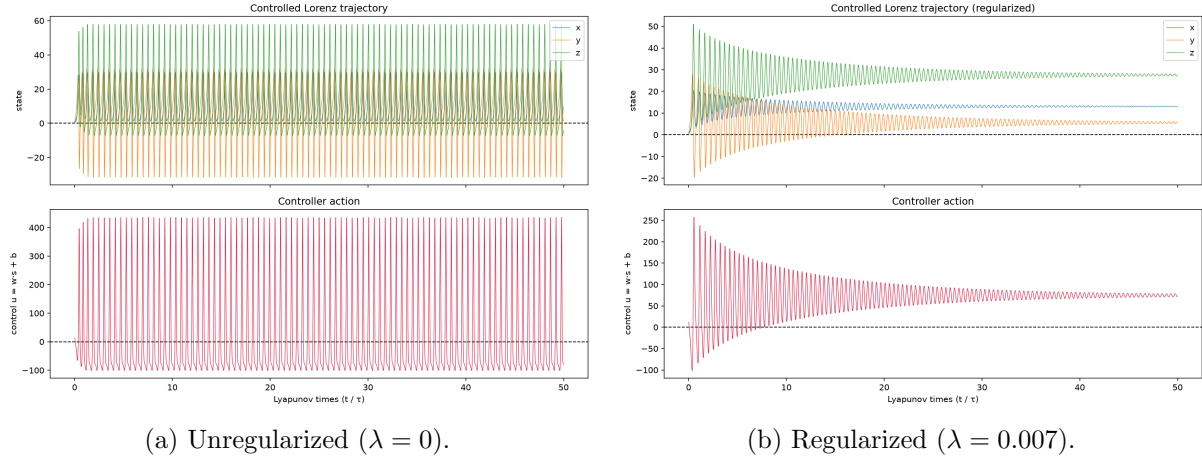


Figure 2: Closed-loop state (top) and control (bottom) over 50 Lyapunov times, for controllers trained on a 1τ window from $(0, 1, 1.05)$. Both keep $x > 0$ for the entire run. The unregularized law does so on a sustained limit cycle demanding $|u|$ up to 435.8; the regularized law spirals into a stable equilibrium and holds it with a constant $u = 74.6$. Note that the two panels do not share a vertical scale.

This is the substantive finding of the regularization experiment, and it is stronger than the usual “the penalty saved energy” claim. The regularized controller is better on *both* terms of the objective: it uses 55% less RMS control *and* attains a task loss $40\times$ lower over the evaluation window (1.1×10^{-3} against 4.4×10^{-2} ; the limit cycle repeatedly carries x down to 1.41, inside the $\varepsilon = 2$ transition band, where φ_ε is appreciably below 1). A penalty term cannot normally improve the objective it is subtracted from. It does here because the unregularized run was never at a task optimum — it was drifting along the flat unbounded valley of Section 7, and the penalty’s real function was to steer it out of that valley and into the basin of a genuine attractor.

9 Numerical results

9.1 Protocol

All runs use initial condition $(0, 1, 1.05)$, Adam with learning rate 0.05, 600 iterations, and full-batch gradients (a single deterministic trajectory). Controllers are initialized at $\theta = 0$, i.e. the free Lorenz system. Training uses a window of L Lyapunov times; evaluation always uses 50 Lyapunov times, so the reported task loss measures *generalization beyond the training horizon*, which is the quantity of interest.

9.2 The barrier in the optimizer

Table 2 is the practical statement of Section 6. At $L = 1$ the optimization succeeds outright: both controllers hold $x > 0$ for 100% of a 50τ evaluation — fifty times the horizon they were trained on. At $L = 2$ and $L = 3$ it fails: the task loss sits at ≈ 0.43 , the trajectory spends only $\approx 57\%$ of its time in the target lobe, and the learned weights bear no resemblance to the

Table 2: Learned controllers as a function of the training window L . $\|u\|_{\text{rms}}$ and the task loss are measured over a 50τ evaluation rollout; $\text{frac}(x > 0)$ is the true (hard) fraction of the evaluation window spent in the right lobe.

L [τ]	λ	$\mathbf{w}^\top$	b	$\ u\ _{\text{rms}}$	$\text{frac}(x > 0)$	$\mathcal{L}_{\text{task}}^{(50\tau)}$
1.0	0	(−1.895, −6.160, 7.150)	+11.935	164.2	1.000	0.0440
1.0	0.007	(−3.168, −6.442, 5.050)	+13.048	80.0	1.000	0.0011
2.0	0	(−1.858, 3.338, −2.630)	+5.314	51.8	0.568	0.4291
2.0	0.007	(−1.698, 3.279, −2.458)	+5.396	49.3	0.569	0.4458
3.0	0	(−0.568, 0.577, −1.874)	−0.355	46.1	0.595	0.4231
3.0	0.007	(−0.602, 0.651, −1.800)	−0.024	43.8	0.588	0.4299

successful ones (note the sign flips in w_2 and w_3 between the $L = 1$ and $L \geq 2$ rows — these are not perturbations of the same solution but entirely different, and much weaker, controllers).

The failure is not one of expressiveness. The $L = 1$ controller lies inside the $L = 2$ hypothesis class and would score $\mathcal{L}_{\text{task}} \approx 0.001$ on the $L = 2$ objective. The optimizer simply cannot find it. By $L = 2$, (23) predicts the gradient is $\sim e^2 \approx 7$ times larger than at $L = 1$ and the landscape has developed structure at scale e^{-2} ; by $L = 3$ the controllers have collapsed toward $\theta \approx 0$ in b and are barely acting at all. This is the signature of an optimizer that has been handed a direction it cannot take a finite step along: it makes no progress and terminates near where it started.

The usable training horizon for this problem is thus between one and two Lyapunov times, and the mechanism — gradient magnitude and landscape roughness both scaling as e^L — is exactly the one derived in Proposition 2 and measured in (25). Note also that regularization does not extend the horizon: at $L = 2$ and $L = 3$ the regularized and unregularized runs fail identically, as Section 7 predicts they must.

9.3 Generalization beyond the training window

The $L = 1$ result is worth isolating. A controller optimized over 110 integration steps holds the system for 5500 steps. This is not luck: by Section 8 the optimizer found a law that creates an attracting equilibrium, and an attracting equilibrium is an infinite-horizon object. Once the closed loop has been made non-chaotic in the region of interest, the horizon over which its behaviour was verified stops mattering. The short window was sufficient because the *structure* the controller must install — a sign change in $\text{Re } \mu$ — is visible within a single Lyapunov time.

10 Discussion

10.1 Summary

Three results:

1. The gradient of a trajectory-averaged loss with respect to controller parameters grows as $e^{t/\tau}$ in the training window, because backpropagation through time accumulates the same tangent propagator whose growth rate *defines* chaos (Proposition 1). Measured: 0.926 nats per Lyapunov time against a predicted 1, over 11.4 decades.
2. The binding horizon is set by optimization, not arithmetic. Double precision permits ≈ 36 Lyapunov times; the optimizer fails between 1 and 2. Adam’s scale invariance hides the divergence without repairing it.

3. An L^2 control-effort penalty restores well-posedness (the unregularized problem has no minimizer, only a minimizing sequence along an unbounded valley) and, at $\lambda = 0.007$, improves both the task loss ($40\times$) and the actuation cost (55%) by steering the closed loop from a large limit cycle into the basin of a stable equilibrium. It does not extend the training horizon, and no penalty on θ can: the barrier belongs to the plant.

10.2 Remedies

The barrier is real but not the end of the subject. Approaches that address it directly:

Truncated BPTT / multiple shooting. Split the window into segments of length $\lesssim \tau$, detach the gradient at the segment boundaries, and sum the per-segment losses. The gradient is then bounded by e^1 regardless of total horizon. The cost is a biased gradient — credit assignment across segment boundaries is discarded — which for a task like lobe confinement, where the relevant causal chain is short, is an acceptable trade.

Least-squares shadowing (LSS). For long-time averages of a chaotic system, the correct object is not $d\langle J \rangle/d\theta$ computed by the naive adjoint — which diverges — but the derivative along a *shadowing* trajectory that remains close to the perturbed attractor. LSS solves a constrained least-squares problem for the shadowing direction and returns a bounded, physically meaningful sensitivity. This is the principled fix when the objective genuinely is a long-time statistic.

Ensemble and zeroth-order methods. Averaging the gradient over an ensemble of initial conditions suppresses the trajectory-specific bursts visible as the staircase in Figure 1, though it does not change the $e^{\lambda_1 T}$ scaling of the ensemble mean. Evolution strategies and other zeroth-order estimators sidestep the propagator entirely at the cost of dimensional scaling — viable here, with $\dim \theta = 4$.

Reformulating the objective. The deepest fix is the one Section 8 points at: the task did not actually require a long horizon. It required installing an eigenvalue sign change, which is a local, short-horizon property. Where a control objective can be restated as a statement about the closed-loop *structure* rather than about a long trajectory, the barrier does not arise.

10.3 Limitations

Every result here is for a single initial condition, a single random seed, a four-parameter affine controller, an unsaturated actuator, and explicit Euler at a fixed step. The exponent (25) is a finite-time estimate along one orbit. The claim that the usable horizon is “between one and two Lyapunov times” is specific to this loss, this optimizer, and this learning rate; the $e^{t/\tau}$ scaling that produces it is not.

A Notation

Symbol	Meaning
$\mathbf{s} = (x, y, z)$	state
σ, ρ, β	Prandtl, Rayleigh, geometric parameter (10, 28, 8/3)
u	scalar control entering the x -equation
$\theta = (\mathbf{w}, b)$	controller parameters, $\mathbf{w} \in \mathbb{R}^3$, $b \in \mathbb{R}$
Δt	integration step, 0.01
λ_1	maximal Lyapunov exponent, 0.9056
$\tau = \lambda_1^{-1}$	Lyapunov time, 1.104
$L = T/\tau$	horizon in Lyapunov times
N	number of Euler steps, $\approx 110 L$
ε	lobe-indicator width, 2
u_{ref}	control normalization scale, 60
λ	effort penalty weight, 0.007
$\mathbf{J}, \tilde{\mathbf{J}}$	open- and closed-loop Jacobians
$\Phi(t, s)$	tangent propagator
a_k	adjoint / costate

B Correspondence with the implementation

Object	Equation	Code
Free Lorenz flow, NumPy	(1), (8)	<code>sim.array_data</code>
Controlled flow, differentiable	(10), (8)	<code>sim.tensor_data</code>
Feedback law	(11)	<code>sim.control_force, sim.create_controller</code>
Lobe indicator	(12)	<code>sim.phi</code>
Task loss	(14)	<code>sim.calculate_loss</code>
Effort penalty	(15)	<code>sim.control_effort</code>
Composite objective, training	(16)	<code>sim.optimize_gradient</code>
Open-loop sensitivity $\partial x(T)/\partial u$	Remark 1	<code>sim.position_gradient</code>
Gradient-growth experiment	Fig. 1	<code>plots.plot_gradient_vs_window</code>
Closed-loop trajectory figures	Fig. 2	<code>plots.plot_control</code>
Attractor coloured by control	—	<code>plots.plot_control_attractor</code>
Sweep over training windows	Tab. 2	<code>l2_plots</code>

Reproduce Figure 1 with `python plots.py -gg 25`, and Figure 2 with `python plots.py -tlt 1 -plt 50 -s` and the same with `-l2`.